

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech. (CL) Examination****March –April, 2018****CL312 Environmental Engineering I****Date: 03.04.2018****Time: 1:30 p.m. To 4:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below: [07]****(1) Groundwater has [01]**

- A.** Higher TDS than surface water
- B.** Lower mineral content than surface waters
- C.** Lower pH values than surface waters
- D.** higher amount of bacteria than surface waters

(2) The most common cause of acidity in water is [01]

- A.** Carbon Dioxide **B.** Oxygen
- C.** Nitrogen **D.** Hydrogen

(3) What do you understand by ‘Detention Time?’ [01]**(4) Draw the layout of surface water treatment plant and mention functions of each unit. [04]****Q - 2(a) List the physical characteristics of water. Explain any two in details. [05]**

(b) Determine the quantity of alum required in order to treat 13 million liters of water per day at a treatment plant, where 12 ppm of alum dose is required. Also determine the amount of carbon dioxide released per liter of water treated. [05]

The chemical reaction:



Molecular weights: Al – 26.97, S – 32.066, O – 16, H – 1.008, C – 12.01.

OR

(b) 2.5 million liters of water per day is passing through a sedimentation tank which is 6 m wide, 15 m long and having a water depth of 3 m. (1) Find the detention time. (2) What is the average flow velocity through the tank? [05]

(c) Describe “Chlorine Demand and Break Point Chlorination” with neat sketch. [04]

- Q - 3(a)** Compare the characteristics of different coagulants used in water treatment. [07]
(b) What is hardness? Give classification of hardness. [07]

OR

- (b)** Sketch and describe in details the working of Pressure filter. [07]

SECTION – II

- Q - 4 Answer the questions:** [07]

- (1)** Define: Environmental Engineering. [01]
(2) For a city developed haphazardly, the layout of distribution pipes preferred, is [01]
A. Ring system **B.** Radial system
C. Grid iron system **D.** Dead end system.
(3) Water supply system includes [01]
A. Digging a well for water **B.** Construction of dams
C. Construction of canals **D.** Entire arrangement from source to distribution.
(4) When the reduced level of the water source is higher than the reduced level of the consumer's place, water is generally supplied [01]
A. By pumping system **B.** By gravitational system
C. Both (a) and (b) **D.** All the above.
(5) Increase in population of a rapidly growing city may be estimated by [01]
A. Arithmetical mean method **B.** Geometrical increase method
C. Incremental increase method **D.** Graphical comparison method
(6) What do you understand by per capita demand? [01]
(7) What is discrete particle settling? [01]

- Q - 5(a)** Explain Environmental ethics in details and explain role of environmental engineer. [05]
(b) Explain salient features of house drainage plan. [06]

OR

- (b)** Differentiate Nahani trap and gully trap showing sketch. [06]
(c) Enlist any three differences between slow sand filter and rapid sand filter. [03]

- Q - 6(a)** What is design period? What are the factors affecting design period. [07]
(b) What is intake structure? Give the classification of the intake structure and explain any one intake structure in details with neat sketch. [07]

OR

- (b)** Explain elevated service reservoir with a neat sketch showing all its components. [07]
